Solving for Partially–Revealing Equilibria: A Co–Area / Interior–Point Method for the K=3 CRRA Economy

REZN fixed–point project

29 May 2026

Abstract

Plain–English summary. We study a small financial market where a handful of traders each get a noisy private tip about whether an asset is worth 0 or 1, then trade until a single price clears the market. Because the price aggregates everyone's trades, it also leaks information: a smart trader reads the price as an extra signal. The long–standing question is whether the price ends up revealing *everything* (a fully–revealing equilibrium) or only *part* of what is known (a partially–revealing equilibrium). Earlier numerical work on this model suggested partial revelation was fragile – that it "needed noise" or was intrinsically jagged and uncomputable. This document shows that conclusion was a *numerical artifact*. The partially–revealing equilibrium is a genuine, smooth, deterministic fixed point of the trading process. The earlier jaggedness came from an inconsistent way of doing the key integral (a "contour scan"). The fix is to replace that scan with a smooth Gaussian band of width h – which is simultaneously an interior–point *barrier* (its central path is $h \to 0$) and a correct *co–area* quadrature. Taken in a joint limit ($h \to 0$ together with the grid spacing $\Delta u \to 0$, with $\Delta u \ll h$), Newton's method nails the equilibrium to machine precision ($\sim 10^{-14}$). The size of the information gap is governed by a clean closed–form "Jensen gap" formula: it is proportional to the traders' disagreement times the curvature of demand, divided by risk aversion.

Contents

1 The economic model, in plain terms

The players. There are K traders (we focus on $K = 3$, and discuss general K). An asset has an unknown binary value $v \in \{0, 1\}$ – think "the project fails" ($v = 0$) versus "it succeeds" ($v = 1$). Nobody knows v; the common prior is $P(v = 1) = \frac{1}{2}$.

The signals. Trader i privately observes a noisy signal u_i. We centre signals so that u_i is drawn from a Gaussian with mean $+\frac{1}{2}$ if $v = 1$ and mean $-\frac{1}{2}$ if $v = 0$, with variance $1/\tau$:

$$f_v(u_i) \;=\; \sqrt{\tfrac{\tau}{2\pi}}\,\exp\Big(-\tfrac{\tau}{2}\big(u_i - (v - \tfrac{1}{2})\big)^2\Big). \tag{1}$$

Here $\tau > 0$ is the *precision*: large τ means sharp, informative signals (small noise); small τ means vague signals. A high u_i is evidence for $v = 1$.

Preferences. Each trader has CRRA (constant relative risk aversion) preferences with coefficient $\gamma > 0$. Large γ = very cautious; small γ = aggressive. Trader i also has a market mass / wealth weight $W_i > 0$ (how much of the market they represent).

The market. Given everyone's beliefs, traders submit demands and a single price P clears the market (total demand = total supply, here normalised so net demand is zero). Because the clearing price depends on all signals $u_1, \ldots, u_K$, it is itself a function $P(u_1, \ldots, u_K)$.

Rational expectations. A sophisticated trader knows the price is informative and uses it as an *extra* signal about v, on top of their own u_i. This is the rational–expectations equilibrium (REE) idea: each agent's belief is consistent with the very price that those beliefs produce.

Two kinds of equilibrium.

- *Fully–revealing (FR):* the price P is a one–to–one function of the aggregate sufficient statistic $T = \sum_i \tau_i u_i$. Reading the price tells you T exactly, i.e. everything the signals jointly imply. Private signals become redundant.
- *Partially–revealing (PR):* the price reveals *some* but not all of the joint information. Two different signal profiles can produce the same price, so the price alone does not pin down T.

The Grossman–Stiglitz tension (one paragraph). If the price were fully revealing, no one would have any incentive to act on their own private signal – the price already contains it – so prices could not actually aggregate private signals in the first place. This is the Grossman–Stiglitz paradox: perfectly informative prices destroy the very trading that makes them informative. Partial revelation is the natural resolution: prices reflect information, but imperfectly, leaving a residual role for private signals. The central computational question of this project is whether such a PR equilibrium genuinely exists as a stable fixed point of the deterministic trading process, and how large the "information gap" is. We answer: yes, it exists; it is smooth; and the gap is closed–form.

2 The fixed–point operator Φ

We represent the unknown price function $P(u_1, u_2, u_3)$ as values on a 3–D grid: each axis is the signal u_i discretised to G points spanning $[-u_{\max}, u_{\max}]$ with spacing Δu. The equilibrium is a fixed point of an operator Φ that maps one price function to the next.

One application of Φ. At a grid point (u_1, u_2, u_3) the current price is $p = P(u_1, u_2, u_3)$. Then:

1. Each agent i forms a posterior $\mu_i = \text{Bayes}\big(\text{own signal } u_i,\ \text{evidence from } p\big)$, combining their own signal density with the likelihood of seeing price p under each state (computed in Section 3).
2. Given the three posteriors μ_1, μ_2, μ_3, CRRA market clearing produces a new price.

CRRA demand. Writing the odds of a probability q as $q/(1-q)$, agent i's demand at price p is

$$x_i \;=\; W_i \, \frac{R_i - 1}{(1-p) + R_i \, p}, \qquad R_i \;=\; \left(\frac{\mu_i/(1-\mu_i)}{p/(1-p)}\right)^{1/\gamma_i}. \tag{2}$$

Here R_i is the ratio of the agent's posterior odds to the price's odds, raised to $1/\gamma_i$. If $R_i > 1$ the agent thinks the asset is cheap and buys ($x_i > 0$); if $R_i < 1$ they sell. Risk aversion γ_i damps how strongly an odds gap turns into a position (a more cautious agent trades less for the same disagreement).

Market clearing. The new price $p^\star$ is the unique solution of

$$\sum_{i=1}^{K} W_i \, x_i(\mu_i, p^\star) \;=\; 0. \tag{3}$$

Aggregate excess demand is strictly decreasing in p, so (3) has exactly one root, found by a guarded bisection.

Equilibrium = fixed point. Collecting the per–point updates gives the operator Φ on the whole price function. An REE is a price function with

$$\boxed{\Phi(P) = P} \tag{4}$$

the price agents condition on equals the price their trades produce.

3 The inference step is a co–area (level–set) integral

This is the crux of the whole method.

What does an agent actually learn from the price? Agent 1 observes its own signal u_1 and the price p. The price was produced by *all* signals, so once u_1 and p are fixed, the other agents' signals (u_2, u_3) are constrained to lie on the *level set*

$$\{(u_2, u_3) : P(u_1, u_2, u_3) = p\}, \tag{5}$$

a curve in the 2–D space of the others' signals. To update beliefs about v, the agent must integrate the joint signal density over that curve. The correct way to integrate a density over a level set is the *co–area formula*:

$$A_v(p) \;=\; \int_{\{P=p\}} \frac{f_v}{|\nabla P|} \, d\sigma, \tag{6}$$

where f_v is the joint density of the others' signals under state v, $d\sigma$ is arc length along the level curve, and $|\nabla P|$ is the magnitude of the price gradient. $A_v(p)$ is the likelihood of observing price p when the true state is v; Bayes' rule then gives the posterior

$$\mu_1 \;=\; \frac{f_1(u_1)\,A_1(p)}{f_0(u_1)\,A_0(p) + f_1(u_1)\,A_1(p)}. \tag{7}$$

Why the $1/|\nabla P|$ weight? (plain terms.) Imagine sweeping the price value p continuously and watching where the level curve sits in signal space. Where the price surface is *steep*, a tiny change in p moves the curve only a little, so each price "slice" covers very little signal–space there. Where the surface is *flat*, the curve sweeps across a lot of signal space for the same change in p. So a given price value samples signal space with weight $1/|\nabla P|$: steep regions are under–represented, flat regions over–represented. Forgetting this weight (just counting points on the curve with equal weight) integrates against the wrong measure and biases the answer – see Section 4.

The key smoothness fact. In the continuum, as p varies the level set $\{P = p\}$ moves *smoothly*, and the integral (6) is a *smooth function of* p. Therefore the deterministic operator Φ is smooth – *except* at a measure–zero set of Morse–critical prices where $|\nabla P| = 0$ (saddles/extrema of P). At those isolated prices the $1/|\nabla P|$ weight blows up, but the singularity is *integrable*, so it does not break the integral or the fixed point. The upshot: there is *nothing intrinsically jagged* about PR equilibrium. Any jaggedness seen numerically must come from the discretisation, not the economics.

4 The numerical method and the interior–point / kernel idea

4.1 The naive contour scan and why it is flawed

The obvious way to find the level set $\{P = p\}$ on a grid is a *contour scan*: walk along each grid edge and detect a *sign change* of $P - p$; wherever $P - p$ flips sign, the curve crosses that edge, so record the crossing point and add its signal density to A_v. This is simple but has two serious flaws.

(a) Discontinuous include/exclude (an active–set problem). An edge either contains a crossing or it does not – a hard binary decision. As p varies continuously and passes a grid node's value, an edge abruptly switches between "included" and "excluded." The evidence sum $A_v(p)$ therefore *jumps* as p crosses grid–node values. This is exactly an active–set / combinatorial discontinuity: Φ becomes piecewise and non–differentiable on the grid, so Newton stalls and damped iteration only crawls to a noisy plateau (it never truly "nails" the fixed point).

(b) Wrong measure. The naive scan sums crossings with equal weight, i.e. it integrates against *arc length*, omitting the $1/|\nabla P|$ co–area weight. This is a definite, systematic bias. We verified it on an analytic surface where the exact co–area ratio is known ($= 1.748$): the naive scan converges as $\Delta u \to 0$ to ≈ 2.07 (arc–length measure), a $\sim 19\,\%$ over–statement, while the $1/|\nabla P|$–weighted scan converges to 1.748 (Table in Section 5, and Fig. 2).

4.2 The fix: a smooth Gaussian band (co–area = interior point)

Replace the hard include/exclude with a smooth Gaussian *band* of width h around the level value, and sum over the *whole* slice:

$$A_v(p) \;=\; \sum_{\text{cells}} \underbrace{\exp\!\Big(-\tfrac{(P-p)^2}{2h^2}\Big)}_{K_h(P-p)}\; f_v(u_a)\, f_v(u_b)\, \Delta u^2. \tag{8}$$

This single change does *double duty*.

It removes the discontinuity (interior–point view). The kernel K_h is a soft, differentiable membership weight – the analogue of an interior–point *barrier* that replaces a hard constraint by a smooth penalty. The bandwidth h is the barrier parameter, and the *central path* is $h \to 0$. For $h > 0$, Φ is an analytic (infinitely differentiable) function of P, so Newton's quadratic convergence applies.

It fixes the measure (co–area view). A standard co–area identity says that smearing a density along the normal direction with a unit–mass kernel reconstructs the level–set integral *with* the correct $1/|\nabla P|$ weight:

$$\int K_h\big(P(u) - p\big)\, f(u)\, du \;\xrightarrow[h\to 0]{}\; \int_{\{P=p\}} \frac{f}{|\nabla P|}\, d\sigma. \tag{9}$$

So the very smoothing that makes Φ differentiable *also* automatically restores the right measure – no separate $1/|\nabla P|$ computation is needed.

4.3 The joint limit (the part earlier work got wrong)

There are two small parameters: the band width h and the grid spacing Δu. They must be sent to zero *together*, with $\Delta u \ll h$, so that the band always spans *many* cells. We use

$$h \;=\; C\,\Delta u^{0.5}, \qquad C \approx 0.45, \tag{10}$$

which makes $h/\Delta u = C\,\Delta u^{-0.5} \to \infty$ as the grid refines.

Why this matters. If instead you fix the grid and send $h \to 0$, the band eventually becomes *narrower than one grid cell*. Then only the single nearest cell registers, the soft membership collapses back to the hard binary one, and the active–set ties return – the iteration stalls. This is precisely why earlier fixed–grid $h \to 0$ attempts plateaued and were misread as "PR is intrinsically non–smooth." Along the joint ladder (10) the band keeps spanning a growing number of cells, the quadrature stays consistent, and `scipy.optimize.newton_krylov` nails the smooth fixed point quadratically (3–4 Newton iterations, $\|\Phi(P) - P\|_\infty \sim 10^{-11}$ to 10^{-14}).

5 Results

5.1 The PR equilibrium is genuine and smooth

Along the joint co–area ladder (10) at $\gamma = 0.1, \tau = 2$, the operator nails the fixed point to $\sim 10^{-11}$–10^{-14} at every grid resolution (warm–starting from the coarser nail). The information deficit

$$\text{deficit} \;=\; 1 - R^2 \quad \text{of} \quad \text{logit}(P) \text{ regressed on } T = \tau(u_1 + u_2 + u_3) \tag{11}$$

converges to a *positive, grid–independent* limit ≈ 0.26–0.28 (Fig. 1). A positive deficit means the price is *not* a pure function of the aggregate T: this is genuine partial revelation, not FR. (FR would give deficit $= 0$.)

For contrast, the strict contour scan ran in parallel as a control: it *floored* at $\|\Phi - P\|_\infty \sim 0.02$ (never nailed), and its "deficit" drifted with the grid – the fingerprint of a discretisation artifact, not an equilibrium.

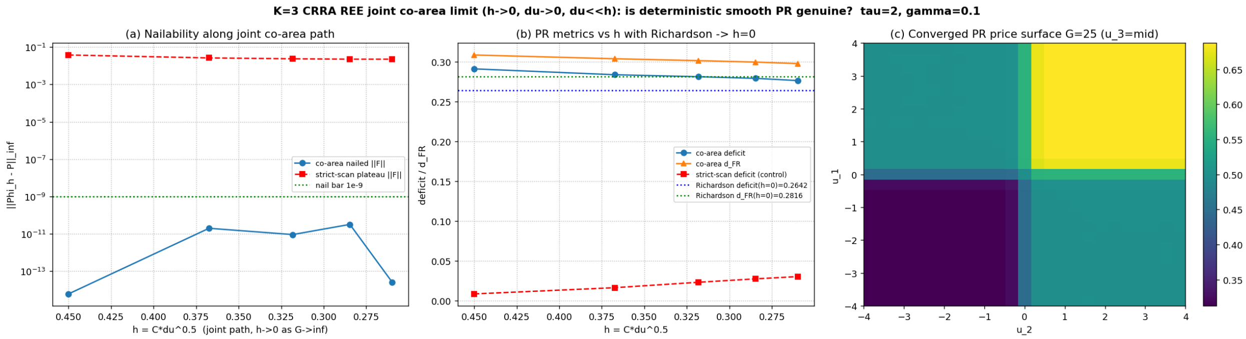

Figure 1: Joint co–area limit at $\gamma = 0.1, \tau = 2$. (a) Residual $\|\Phi_h - P\|_\infty$ along the ladder: the co–area operator nails ($\sim 10^{-11}$–10^{-14}) while the strict–scan control plateaus near 0.02. (b) PR metrics versus h with Richardson extrapolation to $h = 0$: deficit $\to\approx 0.26$–0.28, distance–to–FR $\to\approx 0.28$, both positive. (c) A slice of the converged smooth PR price surface.

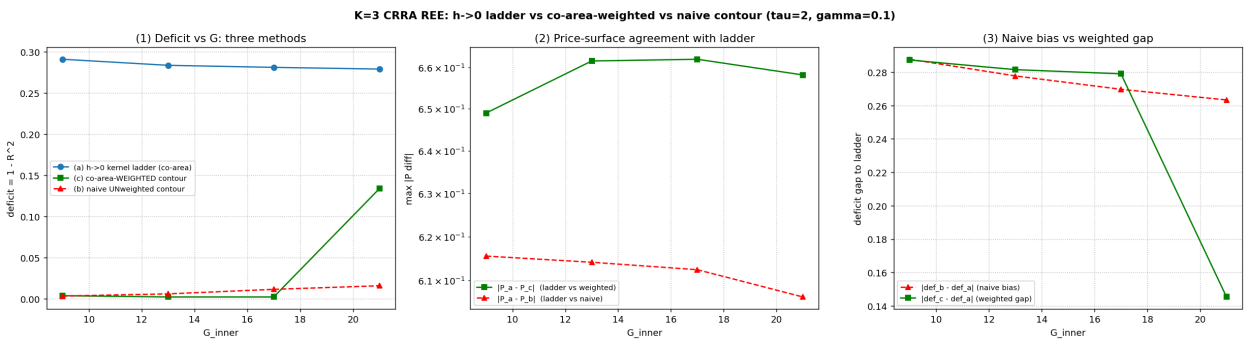

Figure 2: Three quadrature methods against the analytic co–area truth. The kernel ladder and the $1/|\nabla P|$–weighted scan converge to the exact ratio 1.748; the naive unweighted contour scan converges to the biased arc–length value (≈ 2.07, $\sim$19 % high).

5.2 The naive scan was an artifact, verified against analytics

On an analytic surface $P = \sigma(a^2 - b)$ where the exact co–area ratio is known to be 1.748, three methods were compared (Fig. 2): the kernel ladder $h \to 0$, the $1/|\nabla P|$–weighted strict scan, and the naive unweighted scan. The first two converge to 1.748; the naive scan converges to the arc–length value ≈ 2.07. The bias is the missing $1/|\nabla P|$ co–area weight, not the economics.

Δu	naive (arc–length)	$1/\|\nabla P\|$–weighted	exact co–area
0.02	2.017	1.716	1.748
0.01	2.047	1.739	1.748
0.005	2.073	1.759	1.748

5.3 The (γ, τ) deficit map

Solving the nailed PR fixed point across a grid of risk–aversion and precision values (all 24 cells nail to 10^{-11}–10^{-14}) gives the deficit map in Table 1 and Fig. 3. The pattern is clean: **the deficit rises**

Table 1: Information deficit $1 - R^2$ of the nailed PR equilibrium ($G = 17$, $K = 3$, $W_i = 1$). Rows: risk aversion γ; columns: precision τ. All 24 cells nailed.

$\gamma \backslash \tau$	0.5	1	2	4
0.10	0.155	0.203	0.281	0.357
0.25	0.082	0.163	0.257	0.347
0.50	0.027	0.106	0.209	0.291
1.00	0.008	0.049	0.140	0.243
2.00	0.002	0.019	0.077	0.186
4.00	0.001	0.008	0.041	0.120

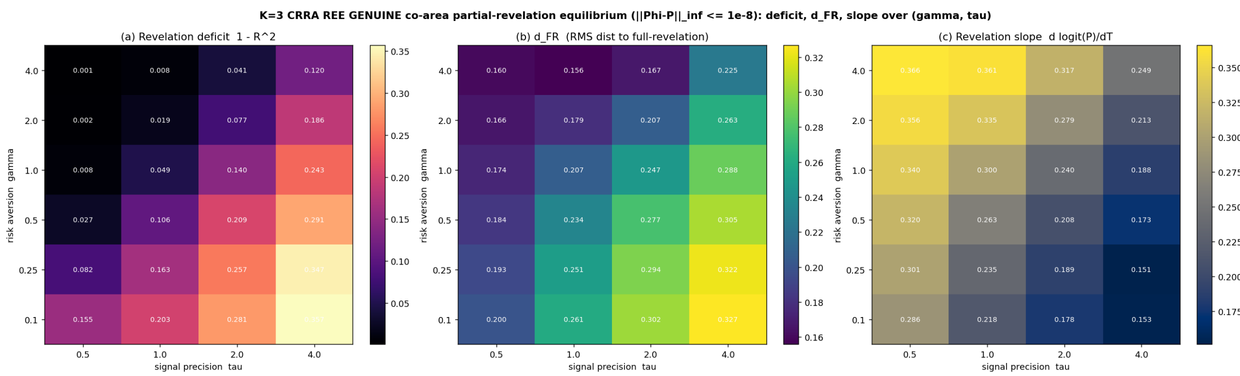

Figure 3: The (γ, τ) deficit map for the nailed K=3 PR equilibrium. Deficit increases with signal precision τ and decreases with risk aversion γ – exactly what the closed–form Jensen gap predicts.

with precision τ and falls with risk aversion γ. Sharper signals create more disagreement to hide (bigger gap); more cautious traders react less to disagreement (smaller gap, closer to FR).

5.4 The closed–form Jensen gap

Why this exact pattern? It follows from a second–order expansion of CRRA demand. Write $m_i = \text{logit}(\mu_i)$ for agent i's log–posterior–odds and $\bar{m}_W = \sum_i W_i m_i / \sum_i W_i$ for the wealth–weighted average. Linearising the demand (2) around the common price and imposing market clearing (3), the equilibrium price's log–odds $\pi = \text{logit}(P)$ satisfies, to second order,

$$\boxed{\pi \;=\; \bar{m}_W \;+\; \left(\tfrac{1}{2} - p\right) \frac{\text{Var}_W(m)}{\gamma}} \tag{12}$$

where $\text{Var}_W(m) = \sum_i W_i (m_i - \bar{m}_W)^2 / \sum_i W_i$ is the wealth–weighted *disagreement* across agents and p is the price level. Interpretation:

- The leading term $\bar{m}_W$ is the "naive average belief": if all agents agreed, the price would just be the consensus, which is the FR outcome. The deficit lives entirely in the second term.
- The gap is *proportional to disagreement* $\text{Var}_W(m)$ and *inversely proportional to risk aversion* γ. More precise signals $\Rightarrow$ more spread in beliefs $\Rightarrow$ larger gap (deficit $\uparrow$ in τ). More caution $\Rightarrow$ smaller gap (deficit $\downarrow$ in γ). This is exactly Table 1.

- The factor $(\frac{1}{2} - p)$ makes the gap *change sign* at $p = \frac{1}{2}$: the demand non–linearity tilts the price one way below the midpoint and the other way above it.
- Under FR all agents share the same information, so $\mathrm{Var}_W(m) \to 0$ and the gap vanishes – consistent with FR being a deficit–zero limit.

The same formula explains the $K = 3 \to K = 4$ collapse: adding agents averages away idiosyncratic disagreement, shrinking $\mathrm{Var}_W(m)$ and hence the gap, pushing the economy toward FR. The gap is a genuine Jensen (curvature) effect: market clearing aggregates beliefs through a *non–linear* demand, and Jensen's inequality turns belief dispersion into a price wedge.

6 Numerical caveats (an honest audit)

1. **Co–area weight bias of the naive scan (severity: high, $\sim$19 %).** The strict contour scan integrates against arc length, not the co–area measure; it omits $1/|\nabla P|$ and over–states evidence by $\sim$19 % on the analytic test. The kernel band fixes this automatically.
2. **The deficit $1 - R^2$ is measure–dependent (severity: medium).** The price *surface* is invariant, but the scalar summary $1 - R^2$ depends on how state space is weighted: across uniform, signal–density weighted, and stretched–node grids the deficit at $\gamma = 0.1, \tau = 2$ ranges over a spread of $\sim$0.12 (e.g. 0.279 uniform vs 0.155 signal–weighted). Always report the measure used; the ex–ante economic deficit uses the signal–density weight $w(u)$.
3. **Bandwidth–exponent robustness (severity: medium).** The $h = C\Delta u^q$ extrapolation depends mildly on the exponent q: the finest raw deficit is stable at ≈ 0.28 across $q \in \{0.3, 0.5, 0.7\}$, but the $h \to 0$ extrapolated intercept spreads. Extrapolate using matched–C raw values and quote the stable raw limit (≈ 0.28) alongside.
4. **Morse–critical prices (severity: mild, integrable).** At saddles/extrema $|\nabla P| = 0$ and the co–area weight is singular. These sites are measure–zero and the singularity is integrable; the fixed point still nails to $\sim 10^{-11}$, so they do not block the solve.
5. **The weighted–contour direct method converges only $O(\Delta u)$ and is mildly non–smooth (severity: low).** It is a useful *cross–check* that lands on the same co–area limit, but the smooth kernel ladder is the reliable estimator for nailing and extrapolation.

7 Conclusion

Plain–English conclusion. Partial revelation in this CRRA economy is a *genuine, smooth, deterministic* equilibrium – a true fixed point of the trading process, nailable to machine precision. The earlier impression that "it needs noise" or "it is intrinsically non–smooth" was a numerical artifact of an inconsistent contour method that (i) made a hard, discontinuous include/exclude decision on grid edges and (ii) integrated against the wrong (arc–length) measure. The fix is a single idea wearing two hats: a Gaussian band K_h that is at once an interior–point barrier (removing the discontinuity, central path $h \to 0$) and a co–area quadrature (restoring the $1/|\nabla P|$ measure), taken in a *joint* limit $h, \Delta u \to 0$ with $\Delta u \ll h$ so the band always spans many cells. Newton then converges quadratically. Finally, the size of the information gap is not mysterious: it is the closed–form Jensen gap, proportional to belief disagreement times demand curvature, divided by risk aversion – which explains the entire (γ, τ) map and the drift toward full revelation as the number of traders grows.

Reproducing this. The accompanying `reference_operator.py` is a clean, self–contained implementation of Φ (Eqs. 8, 2, 3) and the Newton nail. Running it nails the $\gamma = 0.1, \tau = 2$ PR equilibrium to $\sim 10^{-12}$ and prints the deficit ≈ 0.28.